BASIC DEEP LEARNING

Concepts, Neural Networks, Applications, and Future

Presented by [Member Names] · October 25, 2026



What is deep learning?

A branch of artificial intelligence that learns patterns from large amounts of data.

Artificial neural networks recognize images, understand speech, and generate text. A familiar example is face recognition on smartphones.

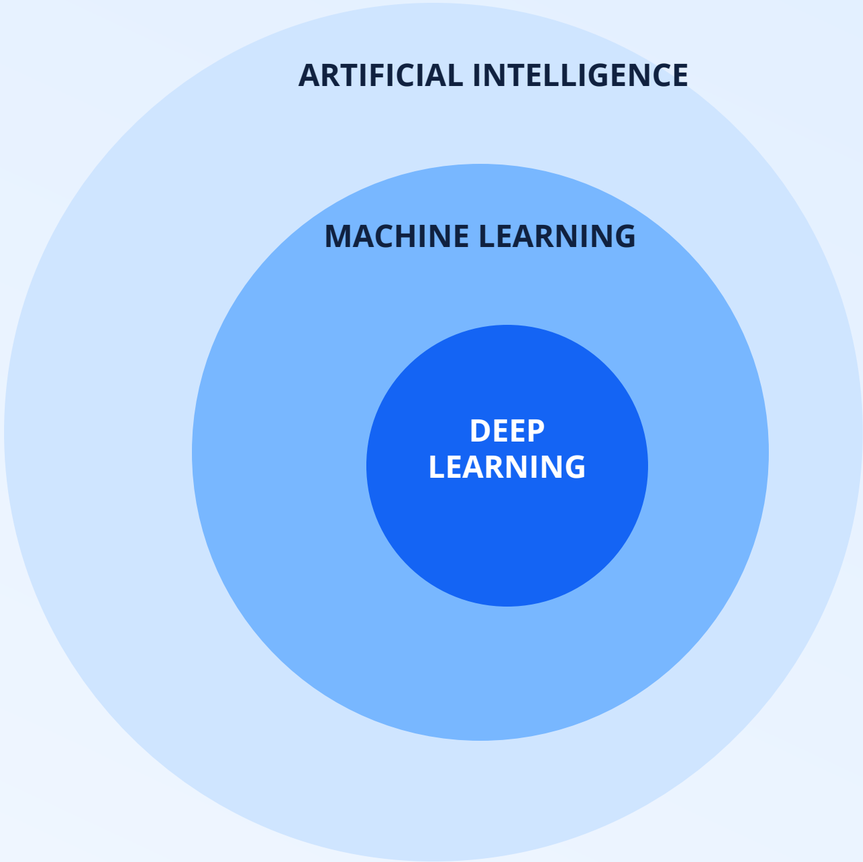
KEY TAKEAWAY

Machines can perform tasks that require human-like intelligence.



One field, nested intelligence

Each inner discipline is a more specialized way to make machines intelligent.



AI plays chess · ML detects spam · DL recognizes faces

How deep learning works

A cat image becomes a prediction — then error turns into learning.

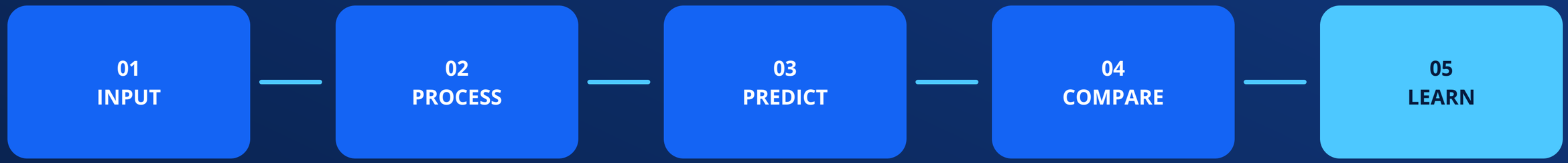


Image or text

Find patterns

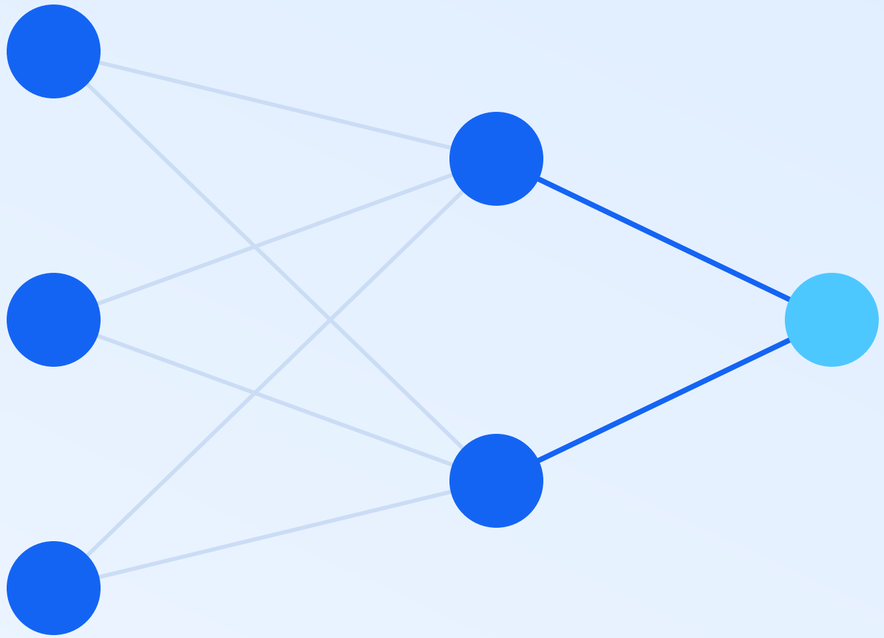
Output a result

Calculate error

Adjust weights

A network of weighted connections

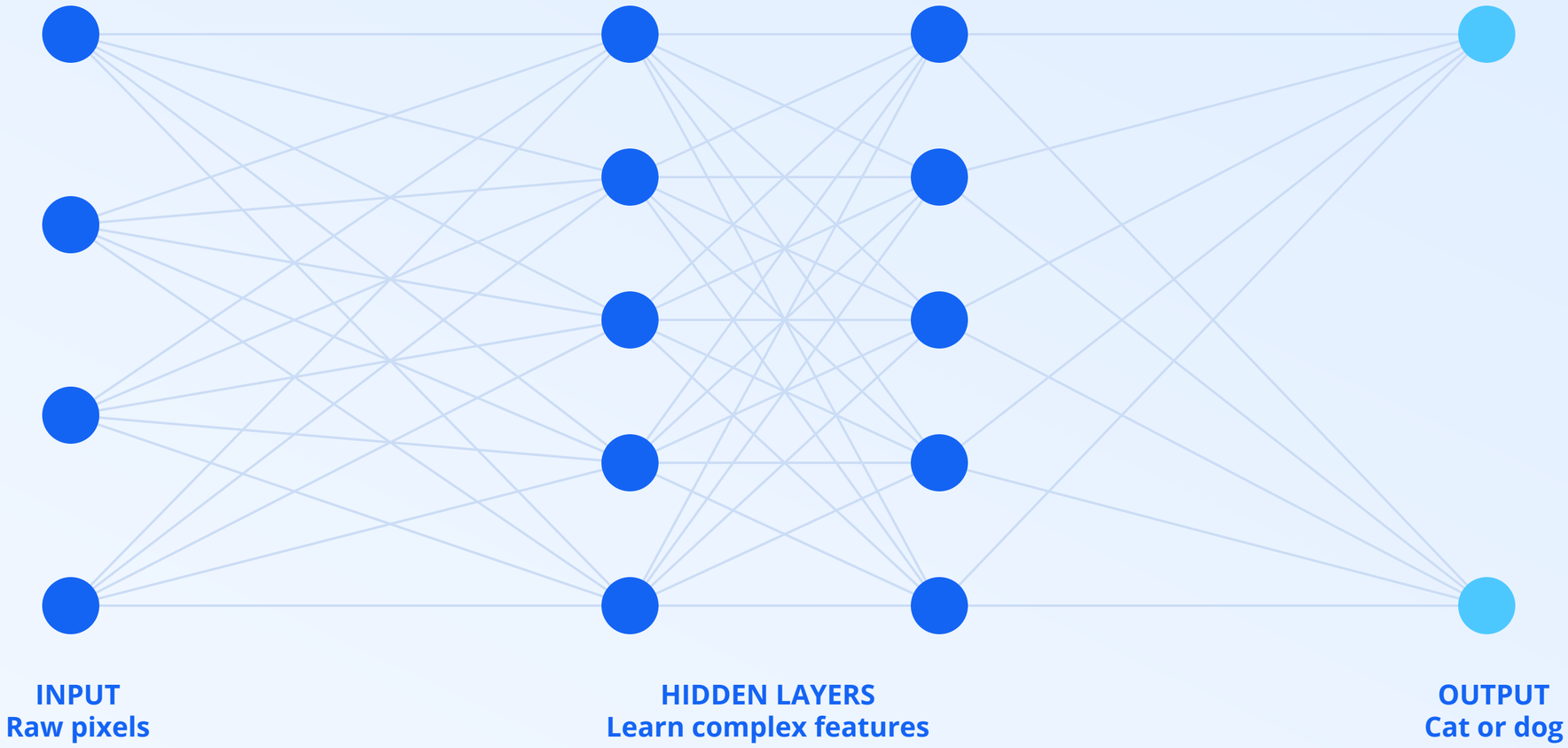
An artificial neural network mimics biological neurons. Interconnected artificial neurons process information through weighted links — powering classification, prediction, and pattern recognition.



FOUNDATION OF DEEP LEARNING

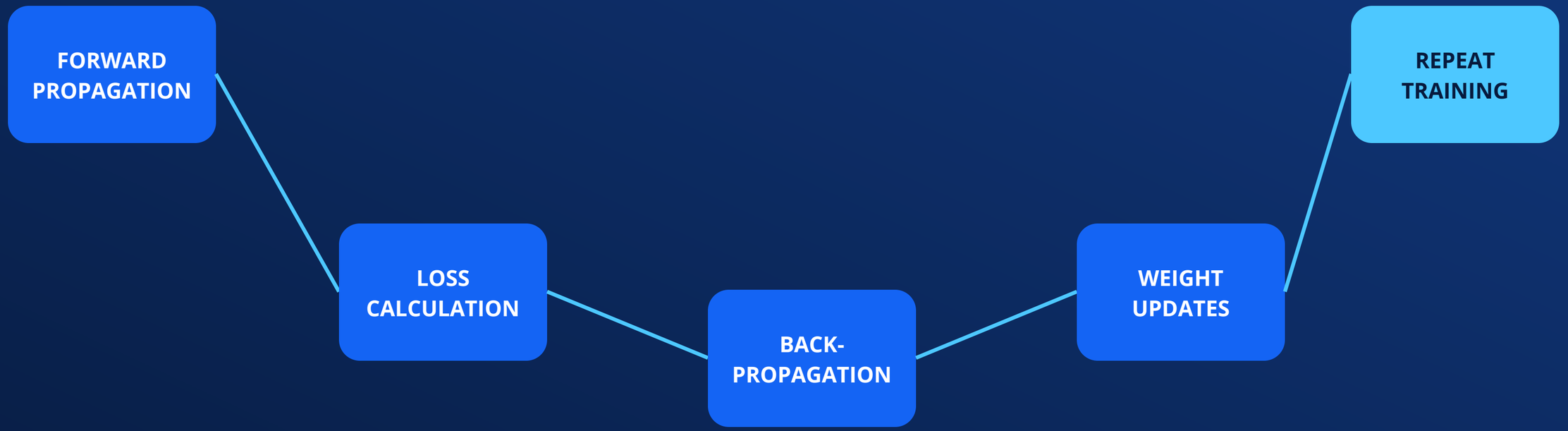
Example: classify an email as spam or not spam.

Layers turn raw data into meaning



Learning is a loop, not a leap

Every example nudges the network toward a better answer.



Example: a dog misclassified as a cat produces error; training adjusts the weights and tries again.

Deep learning is already everywhere



Healthcare

Medical imaging & disease detection



Smartphones

Face unlock, voice & better cameras



Self-driving

Objects, lanes & traffic signs



Generative AI

Text, image, speech, video & code

Why deep learning excels

01

Automatic feature learning

Finds patterns without manual feature design.

02

High performance

Excels in image, speech, and language tasks.

03

Complex data

Works across images, audio, video, and text.

04

Automation

Reduces manual effort on difficult tasks.

05

Improves with training

More data and tuning can lift performance.

Capability comes with trade-offs

LIMITATION	REAL-WORLD IMPLICATION
Large data needs	Limited data availability
High computational cost	Resource constraints
Limited interpretability	Need for explainability
Potential bias	Fairness requirements
Incorrect predictions	Safety concerns

The future is learning forward

HEALTHCARE · EDUCATION · ROBOTICS · TRANSPORTATION · GENERATIVE AI

Deep learning uses neural networks to learn complex patterns — enabling advanced applications while demanding responsible solutions to data, cost, explainability, bias, and safety.

“Deep learning is transforming technology by enabling machines to learn complex patterns.”

“The best way to predict the future is to create it.” — Peter Drucker

THANK YOU!

Any questions?

[Member Names]